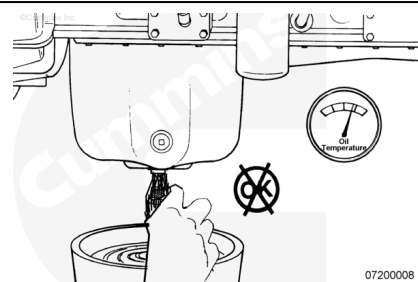


Trainings ▾

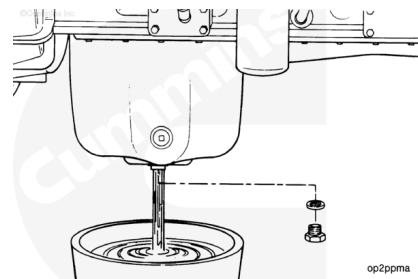
Drain

Change the lubricating oil and filter at the specified oil drain interval. Refer to Procedure 102-002 in Section 2 (</qs3/pubsys2/xml/en/procedures/35/35-102-002-om.html>) of the Operation and Maintenance Manual, ISM and ISMe Engines, Bulletin 3666319.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Note : On standby generator applications, Cummins Inc. recommends oil sampling and analysis at the time of oil change to monitor oil contaminant levels.

Operate the engine until the water temperature reaches 60°C [140°F]. Shut off the engine. Remove the oil drain plug from the bottom of the lubricating oil pan. Do **not** remove the plugs on either side of the oil pan to drain the oil. They will **not** allow the oil to drain completely.

Note : Fittings other than Cummins®-specified parts used in the bottom drain opening of the oil pan must **not** exceed the following size and weight limits:

Oil Drain Fitting Specifications

Length	Diameter	Weight
63.50 mm [2.500 in]	41.28 mm [1.625 in]	0.363 kg [0.80 lb]
Do not use fittings other than the Cummins®-supplied fittings in the side drain location.		

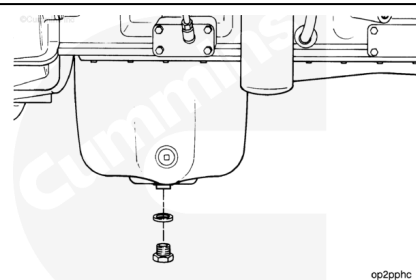
Fill

Clean and check the oil drain plug threads and the seal surface.

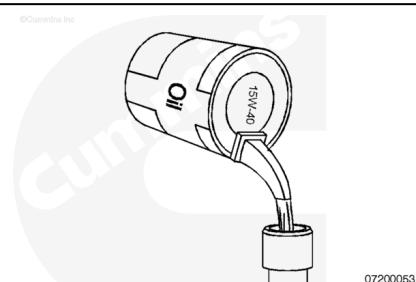
Install the oil drain plug in the lubricating oil pan.

Torque Value: 88 N•m [65 ft-lb]

Check the torque of the side and bottom drain plug to be sure both are tight.



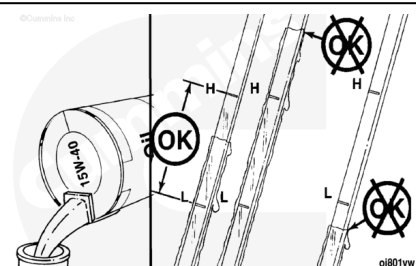
Fill the engine with clean 15W-40 oil. Choose the correct oil for your operating climate. Refer to Procedure 102-002 in Section 2 (/qs3/pubsys2/xml/en/procedures/35/35-102-002-om.html) of the Operation and Maintenance Manual, ISM and ISM^e Engines, Bulletin 3666319.



Fill the engine with clean oil to the H (high) mark on the dipstick. Total system capacity including filter is approximately 39 liters [10.3 gal].

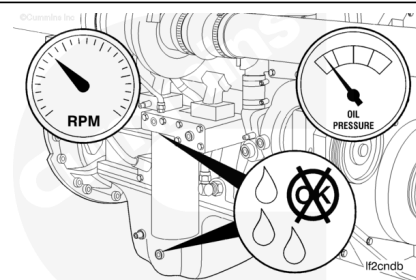
The oil pan capacity is 34 liters [9 gal].

Note : Check Centinel™ makeup tank oil level, if equipped.



⚠ CAUTION ⚠

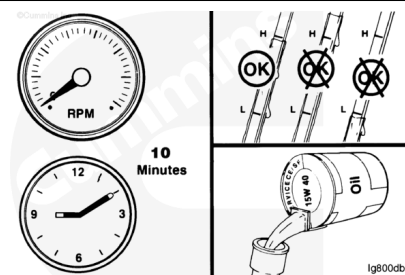
Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level in the oil pan.



Operate the engine at idle speed to inspect for leaks at the filter(s) and the drain plug.

Shut off the engine. Wait approximately 10 minutes for the oil to drain back from the upper parts of the engine to the oil pan.

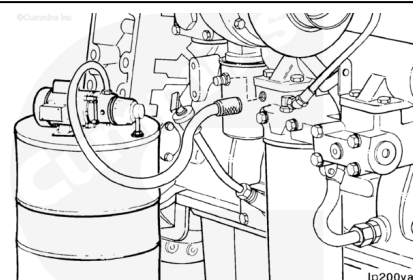
Check the oil level again. Add oil as necessary to bring the level up to the H (high) mark on the dipstick.



Prime

⚠ CAUTION ⚠

The lubricating oil system must be primed before operating the engine after a rebuild or extended shutdown to avoid internal component damage. If an external pressure pump is used, do not prime the system after the filter. The filter will be damaged.



Remove the plug at the "filter-in" fitting of the lubricating oil filter head. This will allow oil to flow in the correct direction through the filter.

Install an oil supply line connected to a source of clean 15W-40 lubricating oil in the Compuchek® fitting drilling of the filter head.

Prime the lubricating system until the oil pressure gauge indicates a positive pressure.

The oil pressure **must not** exceed the maximum of 276 kPa [40 psi].

Remove the oil supply and install the Compuchek® fitting.

Torque Value: 54 n•m [40 ft-lb]

